

M CHATURVED TEJAS

AI/ML Engineer — Generative AI Engineer — NLP & Intelligent Systems

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LinkedIn — GitHub

SUMMARY

AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, and Embedded Systems. Skilled in building end-to-end AI applications, real-time analytics systems, conversational AI platforms, and intelligent automation solutions using Python, TensorFlow, PyTorch, and modern AI frameworks. Experienced in model optimization, API integration, full stack AI development, inference pipelines, and scalable deployment workflows.

EXPERIENCE

AI/ML Intern

ParkNSecure Pvt Ltd

06/2025 – 07/2025

- Developed NLP-based chatbot automation system for smart parking management using Python and conversational AI techniques
- Implemented contextual intent recognition, geospatial query handling, and real-time response generation workflows
- Optimized inference pipelines and API communication using caching mechanisms to reduce response latency
- Improved user query resolution efficiency through intelligent conversational workflow integration

AI/ML Intern

CodTech IT Solutions

07/2024 – 09/2024

- Performed data preprocessing, feature engineering, and class imbalance handling using SMOTE techniques
- Evaluated machine learning models using cross-validation, ROC-AUC, precision-recall, and performance evaluation metrics
- Improved model generalization through hyperparameter tuning and optimization strategies
- Worked on predictive analytics workflows and machine learning pipeline enhancement

EDUCATION

Bachelor of Technology in Computer Science and Engineering (AI & ML)

GITAM University

08/2022 – 05/2026

CGPA: 8.17/10

PROJECTS

CETRI AI — Cognitive Enhanced Technical Response Intelligence

2026

- Developed full stack AI assistant platform integrating conversational AI, voice interaction, automation, memory handling, and real-time communication systems
- Built desktop and web-based AI ecosystem using Python, FastAPI, Next.js, React, TypeScript, WebSockets, Whisper, and Tailwind CSS
- Improved interaction efficiency by ~87% through asynchronous processing, backend optimization, and low-latency communication workflows
- Designed unified productivity platform combining AI chat, coding assistance, research workflows, automation tools, and system monitoring

NeuroVision AI — Intelligent Surveillance Analytics Platform

2026

- Architected computer vision surveillance platform for anomaly detection, crowd analytics, object tracking, and real-time event monitoring across multi-camera environments
- Implemented deep learning pipelines using YOLO, OpenCV, PyTorch, FastAPI, PostgreSQL, WebSockets, and React-based dashboards

- Enabled low-latency object detection, pose estimation, trajectory prediction, and event-driven incident reporting workflows
- Designed centralized monitoring infrastructure for smart campuses, malls, parking systems, and enterprise surveillance environments

AI Smart Parking Chatbot — LLM Integration

01/2025 – 03/2026

- Engineered smart parking assistance system to address vehicle parking congestion and navigation challenges in metropolitan environments
- Built conversational AI chatbot using Gemini API, Flask, OpenCV, PyTorch, YOLOv5, and NLP-driven inference pipelines
- Achieved ~80% response accuracy and reduced parking query resolution time by ~50% through prompt engineering and API workflow optimization
- Designed deployment architecture for colleges, hospitals, malls, temples, and public smart parking infrastructures

Handwriting Analysis for Behavioral Insight Prediction

05/2025 – 03/2026

- Developed deep learning-based handwriting analytics platform for behavioral pattern analysis and DASS-aligned emotional severity prediction
- Implemented predictive ML pipelines using PyTorch, Streamlit, OpenCV, scikit-learn, and CNN-based feature extraction techniques
- Achieved ~87% classification accuracy through feature engineering, hyperparameter tuning, and model optimization
- Designed research-oriented assessment platform for behavioral analytics and counselor pre-screening support

Smart Safety System — AI-Based Compliance Monitoring

01/2024 – 04/2024

- Developed computer vision-based compliance monitoring system for automated face mask detection and policy violation analytics
- Built facial analytics pipeline using MTCNN, MobileNetV3, TensorFlow, DeepFace, FaceNet512, and OpenCV
- Reduced false-positive alerts by ~35% through model fine-tuning and threshold optimization
- Designed monitoring solution for healthcare facilities, laboratories, campuses, and controlled-access environments

Autonomous Obstacle Detection Robot

2024

- Engineered autonomous robotic navigation prototype inspired by self-driving vehicle technologies for obstacle avoidance and adaptive path navigation
- Integrated Arduino UNO, HC-SR04 ultrasonic sensors, motor drivers, DC motors, and Embedded C programming for real-time mobility control
- Achieved stable obstacle detection within 6–8 meter sensing range across indoor and outdoor environments
- Designed IoT-enabled robotics system for smart navigation and autonomous mobility experimentation

SKILLS

Programming Languages: Python, Java, SQL, Embedded C

AI & Machine Learning: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, CNNs, Predictive Analytics, Feature Engineering, Prompt Engineering, Model Optimization

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, Streamlit

Web & Backend Technologies: FastAPI, REST APIs, WebSockets, Flask, React, Next.js, Tailwind CSS

Tools & Platforms: Git, Docker, Jupyter Notebook, Arduino UNO

Cloud & Technologies: AWS Cloud Foundations, Microsoft Azure AI, Google Cloud AI, API Integration

Embedded & Robotics: Embedded Systems, Robotics, Sensor Integration, IoT Fundamentals, Automation Systems

CERTIFICATIONS

- Google Cloud AI Certification Badges
- AWS Cloud Foundations
- Microsoft Azure AI Fundamentals
- Accenture Virtual Internship

KEY ACHIEVEMENTS

- Solved 350+ competitive programming problems across coding platforms
- Participated in AI, Machine Learning, and software development hackathons
- Developed multiple AI-driven applications focused on automation, analytics, and real-world problem solving

LANGUAGES

English (Proficient) Telugu (Native) Hindi (Advanced) Spanish (Intermediate) German (Intermediate)

INTERESTS

AI Research — Robotics — Intelligent Automation Systems
NSS Volunteer — Community Service